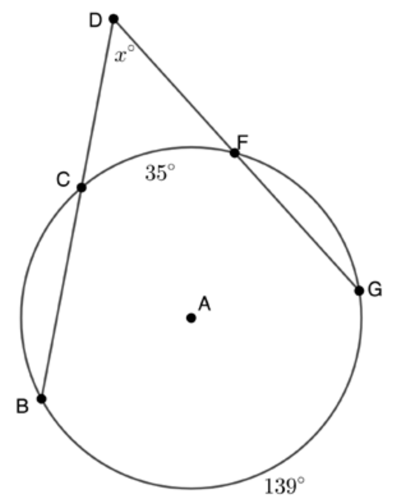
Geometry
Unit 10
Lesson 8 Practice

Name____________________
Date____________________
Period__________________

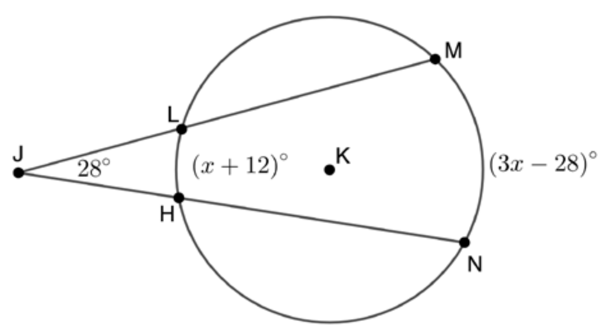
For questions 1-2, Circle A has points B, C, F, and G on the circle. Secant lines $\overline{BD}$ and $\overline{GD}$ intersect at point D outside the circle. The $m\widehat{BG} = 139°$, $m\widehat{CF} = 35°$, and $m\angle CDF = x°$.



1. Write a formula that can be used to find the value of x.

2. Find the value of x.

For questions 3-5, Circle K has points N, H, L, and M on the circle. Secant lines $\overline{NJ}$ and $\overline{MJ}$ intersect at point J outside the circle. The $m\widehat{LH} = (x + 12)°$, $m\widehat{MN} = (3x - 28)°$, and $m\angle HJL = 28°$.



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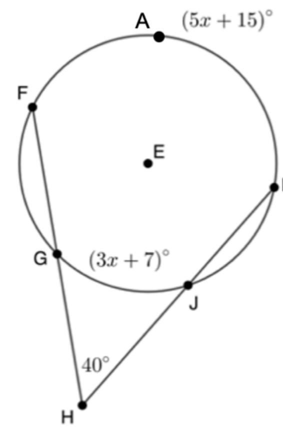
3. Find the value of x.

4. Find the $m\widehat{MN}$.

5. Find the $m\widehat{LH}$.

For questions 6-8, Circle E has points F, G, J, and K on the circle. Secant lines $\overline{FH}$ and $\overline{KH}$ intersect at point H outside the circle. The $m\widehat{FAK} = (5x + 15)°$, $m\widehat{GJ} = (3x + 7)°$, and $m\angle GHJ = 40°$.



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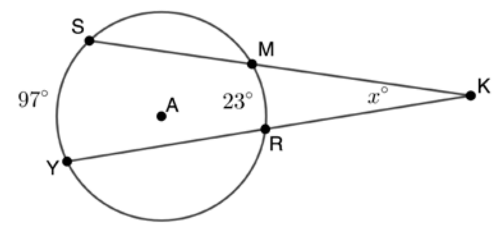
6. Find the value of x.

7. Find the $m\widehat{GJ}$.

8. Find the $m\widehat{FAK}$.

For questions 9-10, Circle A has points S, M, Y, and R on the circle. Secant lines $\overline{SK}$ and $\overline{YK}$ intersect at point K outside the circle. The $m\widehat{SY} = 97°$, $m\widehat{MR} = 23°$, and $m\angle MKR = x°$.



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9. Write a formula that can be used to find the value of x.

10. Find the value of x.